

L-theanine intervention enhances human gammadeltaT lymphocyte function.

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Human gammadeltaT lymphocytes are a subset of T cells and are a first line of defense against microbes and tumors. These gammadeltaT cells can be primed by nitrogen-containing bisphosphonates, and certain short-chain alkylamines. These primed gammadeltaT cells have an enhanced capacity to proliferate and to secrete cytokines upon ex vivo exposure to a wide variety of microbes and tumor cells. The largest dietary source of alkylamines is L-theanine, an amino acid unique to tea beverages that is catabolized to ethylamine. Supplementation of subjects with capsules containing L-theanine and catechins has recently been shown to decrease the incidence of cold and flu symptoms, while enhancing gammadeltaT cell function.